INCH POUND
MIL-PRF-1/588B
29 January 1999
SUPERSEDING
MIL-E-1/588A
18 December 1974

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBE, CATHODE RAY
TYPE 2AP1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE-RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:	V	V dc	v	V dc	V dc	fL	Meg	Meg	V dc	ft
Maximum:	6.9	0	660	550	1,100	- - -	1.5	1.0	-125	10,000
Minimum:	5.7	-125	- - -	- - -	500	2.0	- - -	- - -	- - -	- - -
Test conditions:	6.3	Adj	- - -	Focus	1,000	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification - Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Qualification inspection							
Neck and bulb alignment (electrostatic types)	5101	- - -		Diam	- - -	1.63	Inches
Cathode illumination	5216	- - -		- - -	- - -	- - -	- - -
Deflection factor uniformity	5248	- - -		- - -	- - -	- - -	- - -
Direct-interelectrode capacitance	1331	- - -	Control grid to all Cathode to all D1 to D2 D3 to D4 D1 to all D3 to all D1 to all, except D2 D2 to all, except D1 D3 to all, except D4 D4 to all, except D3	Cg1 Ck C1D2 C3D4 CD1 CD3 CD1 CD2 CD3 CD4	- -	11.0 7.5 2.0 2.0 12.5 13.0 12.0 12.0 10.0 9.0	pF pF pF pF pF pF pF pF pF pF
Pressure (implosion)	1141	- - -		- - -	- - -	- - -	- - -
Vibration	5111	- - -		Width	- - -	1.0	mm
Base material insulating quality	1216	- - -	Zone 5 (minimum)	- - -	- - -	- - -	- - -
Conformance inspection, part 1							
Voltage breakdown	5201	- - -		- - -	- - -	- - -	- - -
Voltage breakdown (electrostatic types)	5201	- - -		- - -	- - -	- - -	- - -
Gas “cross”	5206	1/	Light = 2 fL	- - -	- - -	- - -	- - -
Bulb, screen and faceplate quality	5106	- - -		- - -	- - -	- - -	- - -
Light output	5221	1/		Light	2.0	- - -	fL
Modulation	5223	- - -	Light = 2 fL	ΔEc	- - -	50	V dc
Spot position (electrostatic deflection)	5231	- - -		- - -	- - -	10	mm
Spot displacement (leakage)	5231	- - -		Displ	- - -	5	mm
Grid cutoff voltage	5241	- - -		Ec1	-30	-90	V dc
Grid No.1 leakage current	5251	- - -		- - -	- - -	- - -	- - -
Anode No. 2 leakage current	5251	- - -		- - -	- - -	- - -	- - -

See footnotes at end of table.

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TABLE I. Testing and Inspection -Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits Min	Limits Max	Units
Conformance inspection, part 2							
Heater current	1301	- - -		If	540	660	mA
Electrode current (anode 1)	5201	- - -	Ec1 = 0	Ib1	-50	10	μA dc
Electrode current (cathode)	5201	- - -	Light = 2 fL	Ik	- - -	1,000	μA dc
Base alignment (electrostatic types)	5101	- - -	+ 3D4, pin No. 1	- - -	- - -	- - -	- - -
Angle between traces	5101	- - -		Angle	86	94	Degrees
Neck and base alignment (electrostatic types)	5101	- - -		- - -	- - -	- - -	- - -
Stray light emission (conventional types)	5216	- - -		- - -	- - -	- - -	- - -
Line width “A” (electrostatic deflection)	5226	2/	Light = 2 fL	Width	- - -	0.45	mm
Line width “B” (electrostatic deflection)	5226	3/	Light = 2 fL	Width	- - -	0.55	mm
Focusing voltage at cutoff	5246	- - -		Eb1	200	300	V dc
Focusing voltage (zero-bias)	5246	- - -		Eb1	140	300	V dc
Deflection factor	5248	- - -	1D2	DF	195	265	V dc/in.
Deflection factor	5248	- - -	3D4	DF	167	225	V dc/in.
Heater-cathode leakage current	5251	- - -		- - -	- - -	- - -	- - -
Secureness of base, cap or insert	1101	- - -		- - -	- - -	- - -	- - -
Base pin solder depth	1111	- - -		- - -	- - -	- - -	- - -
Permanence of marking	1105	- - -		- - -	- - -	- - -	- - -
Conformance inspection, part 3							
Life test	- - -	- - -	Group B; Light = 2 fL; Eb2 = 1,100 V dc; t = 500 hours	- - -	- - -	- - -	- - -
Life-test end points:							
Line width “A”	5226	- - -	Light = 2 fL	Width	- - -	0.50	mm
Line width “B”	5226	- - -	Light = 2 fL	Width	- - -	0.65	mm
Modulation	5223	- - -	Light = 2 fL	ΔEc	- - -	50.0	V dc

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TABLE I. Testing and inspection -Continued.

NOTES:

1/ This test to be performed at the conclusion of the holding period.

2/ The deflection plates shall be returned to anode No. 2 through a minimum of 2.5 megohm resistors. The light output shall be set at 2 fL. The high frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction of high frequency scanning. Readjustment shall be made for best overall focus. The tubes shall be observed for deflection defocusing, astigmatism and spot ellipticity observable to the eye, as evidenced by fuzziness due to lack of sharpness of trace (usually around the edges), bow-tying (irregular widths of any line when observed at different points) and bowing of trace other than that normally caused by curvature of the bulb. This test for focus is to be made in addition to the line width measurement.

3/ The same conditions shall be set up as described in 2/ except that the connection of deflecting elements to the low and high frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from 2/ conditions. An examination for defocusing, astigmatism or spot ellipticity shall be made as in 2/.

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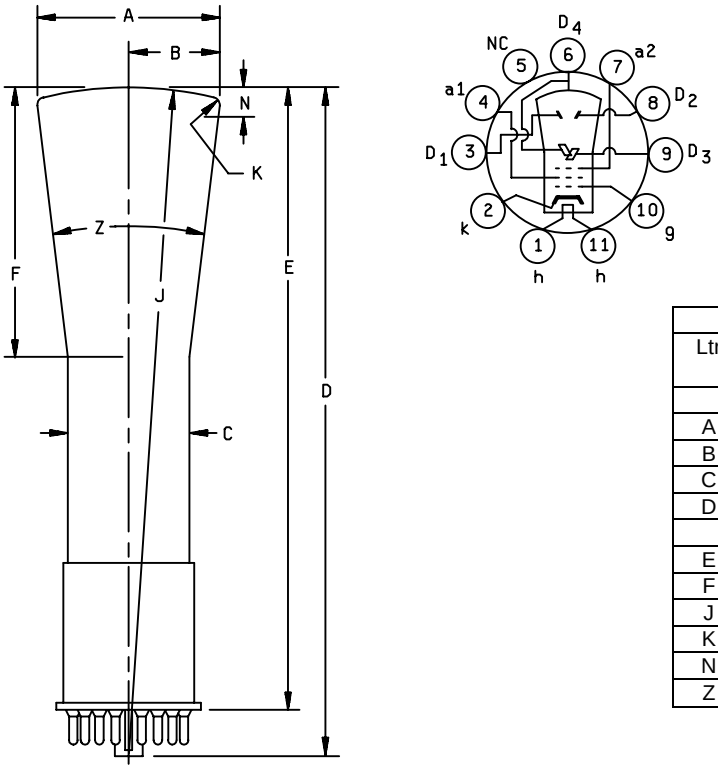

Dimensions				
Ltr	Inches		Millimeter	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	1.937	2.063	49.20	52.40
B	Note 1			
C	1.312	1.438	33.32	36.53
D	7.250	7.626	184.15	193.70
Reference dimensions (see note 4)				
E	6.875		174.63	
F	3.062		77.77	
J	8.000 (203.20) R			
K	.188 (4.78) R			
N	.225 (5.72)			
Z	12° 37’			

NOTES:
1. The minimum useful screen radius shall not be less than .875 inches (22.23 mm).
2. The base shall be a large wafer magnal 11-pin, with flared sleeve, (B11-36) type or a small magnal 11-pin (B11-33) type.
3. The bulb shall be a J16A type.
4. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing for electron tube type 2AP1A.

Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Navy - MC

Preparing activity:
DLA - CC

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